

All Numbered Kozma–Nitzan Conjectures and Questions: a Source-Set Proof and Lean Formalization Blueprint

Prepared from the supplied notes and formal development

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Result. All five numbered conjectures in Kozma–Nitzan—Conjectures 1, 2, 3, 4, and 6—and all four numbered questions—Questions 5, 7, 8, and 9—follow from a guarded, source-set extension of the conditioned slack hierarchy. That extension is proved below from the same finite-product and BHK inputs used by the supplied development. In fact we prove stronger statements. In Conjecture 4 one may use the particular relay minimizing the *unconditional* cluster mean, which answers Question 7. A finite-dimensional separation argument gives the common coefficients requested in Question 5. The strong pre-FKG theorem answers Question 8, and conditioning on the star of the observer answers Question 9. Finally, Conjecture 6 remains true without either of its two endpoint assumptions. The central device is to allow a finite set of vertices, rather than one vertex, to be the source in the avoided first-relay inequality.

The full source-set wrapper described here has not yet been submitted to a Lean kernel. The probabilistic input it needs is already present in the repository, including set clusters, set markers, two-set conditional association, and the CSH peeling theorem. Section 10 separates the new declarations into small formalization blocks.

Abstract

Let independent bond percolation be given on a finite weighted graph. For a finite source set S , write $C_S = \bigcup_{s \in S} C_s$. We prove a source-set, avoided version of the first-relay inequality. If Y is an avoided set, T is a relay set, Φ is increasing, and the relays are ordered by the conditional means

$$m_x^Y = \mathbb{E}[\Phi(C_x) \mid x \not\leftrightarrow Y],$$

then

$$\mathbb{E}[\Phi(C_S); S \not\leftrightarrow Y, S \leftrightarrow T] \geq \sum_{x \in T} \mathbb{P}(J_x^{S,Y}) m_x^Y,$$

where $J_x^{S,Y}$ says that x is the first relay met by C_S . The proof is the CSH surplus-transfer and relay-peeling proof with a set-valued source. We give all of the quantities and the two inequalities used in the induction.

A monotone projection then shows that if $a \in A$ minimizes $\mathbb{E}F(C_a)$, it is the particular relay which works in Conjecture 4. Taking $F(K) = \mathbf{1}\{b \in K\}$ proves the strong pre-FKG inequality and hence Conjectures 2, 1, and 3. Finally take the source set $S = \{v, w\}$. The union $C_v \cup C_w$, its connection events, and its avoidance events are unchanged by the state of the internal edge $e = \{v, w\}$. The avoided source-set inequality therefore transfers verbatim to the graph in which e is forced open, proving Conjecture 6 with the old minimizer a .

The same fixed-minimizer theorem resolves the remaining numbered questions. Convex separation produces one probability vector on A which works simultaneously for every target b (Question 5). Questions 7 and 8 are respectively the fixed-minimizer and minimum forms of the

strong pre-FKG inequality. For Question 9, reveal all pairs incident to the observer; their open endpoints form a random source set in the graph with the observer’s star deleted, to which the source-set theorem applies configuration by configuration. An elementary finite-edge exhaustion extends the needed inequality to countable graphs; combined with Kozma–Nitzan’s Theorem 6, this also gives $\mathbb{P}_{p_c}(|C(0)| = \infty) = 0$ on $\mathbb{Z}^d$ for every $d \geq 2$.

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1 Model and exact targets

Let V be a finite vertex set and let every unordered pair e be open, independently, with probability $p_e \in [0, 1]$. A pair of weight zero is simply an absent edge. The open cluster of x is C_x . For a finite source set $S \subseteq V$, put

$$C_S := \bigcup_{s \in S} C_s. \quad (1)$$

Thus $S \leftrightarrow B$ means $C_S \cap B \neq \emptyset$, and $S \nleftrightarrow B$ means $C_S \cap B = \emptyset$. We use the same symbols when one of the sets is a singleton. All sets in this document are finite.

A cluster functional is an increasing map $F : 2^V \rightarrow \mathbb{R}$. The paper’s slightly more intrinsic definition of a monotone cluster property is equivalent on the support of a finite percolation measure: condition (2) in that definition makes the value common to all roots of one cluster, and an isotone function on the poset of realizable clusters extends to 2^V by

$$F(K) = \max\{f(L) : L \text{ is a realizable cluster and } L \subseteq K\}.$$

An irrelevant constant can be inserted when the displayed set is empty.

We record the five numbered conjectures in the form used below. We always assume $A \neq \emptyset$.

Conjecture 1.1 (Kozma–Nitzan 1). For vertices o, b and a relay set A ,

$$\mathbb{P}(o \leftrightarrow b) \geq \mathbb{P}(o \leftrightarrow A) \min_{x \in A} \mathbb{P}(x \leftrightarrow b). \quad (2)$$

Conjecture 1.2 (Kozma–Nitzan 2). One has

$$\mathbb{P}(o \leftrightarrow b) \geq \min_{x \in A} \mathbb{P}(o \leftrightarrow A, x \leftrightarrow b). \quad (3)$$

The useful stronger form has $\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A)$ on the left.

Conjecture 1.3 (Kozma–Nitzan 3). For every $\varepsilon > 0$ there is a $\delta > 0$, independent of the graph and of $|A|$, such that

$$\mathbb{P}(o \leftrightarrow A) > 1 - \delta, \quad \mathbb{P}(x \leftrightarrow b) > 1 - \delta \quad (x \in A)$$

imply $\mathbb{P}(o \leftrightarrow b) > 1 - \varepsilon$.

Conjecture 1.4 (Kozma–Nitzan 4). For every increasing cluster functional F ,

$$\mathbb{E}[F(C_o); o \leftrightarrow A] \geq \min_{x \in A} \mathbb{E}[F(C_x); o \leftrightarrow A]. \quad (4)$$

For $e = \{v, w\}$, write G/e for the weighting obtained by replacing p_e by 1. This is equivalent to contraction for connectivity events.

Conjecture 1.5 (Kozma–Nitzan 6). Suppose $a \in A$ minimizes $\mathbb{P}_G(x \leftrightarrow b)$ over $x \in A$ and

$$\mathbb{P}_G(x \leftrightarrow b) \geq \mathbb{P}_G(x \leftrightarrow A) \mathbb{P}_G(a \leftrightarrow b), \quad x \in \{v, w\}.$$

Then

$$\mathbb{P}_{G/e}(v \leftrightarrow b) \geq \mathbb{P}_{G/e}(v \leftrightarrow A) \mathbb{P}_{G/e}(a \leftrightarrow b). \quad (5)$$

Conjecture 1.6 (Kozma–Nitzan Question 5). Do there exist coefficients $c_x \geq 0$, depending on G, o, A but not on b , with $\sum_{x \in A} c_x = 1$, such that simultaneously for every $b \in V$,

$$\mathbb{P}(o \leftrightarrow b) \geq \sum_{x \in A} c_x \mathbb{P}(o \leftrightarrow A, x \leftrightarrow b)? \quad (6)$$

The remaining candidate questions all ask whether a specified relay a satisfies

$$\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) \geq \mathbb{P}(o \leftrightarrow A, a \leftrightarrow b). \quad (7)$$

Conjecture 1.7 (Kozma–Nitzan Question 7). Does (7) hold when a minimizes $\mathbb{P}(x \leftrightarrow b)$ over $x \in A$?

Conjecture 1.8 (Kozma–Nitzan Question 8). Does (7) hold when a minimizes $\mathbb{P}(o \leftrightarrow A, x \leftrightarrow b)$ over $x \in A$?

Conjecture 1.9 (Kozma–Nitzan Question 9). Let H be obtained from G by deleting every pair incident to o . Does (7), with probabilities evaluated in G , hold when a minimizes $\mathbb{P}_H(x \leftrightarrow b)$ over $x \in A$?

There is no Conjecture 5 and no Question 6 in the paper. Its complete numbered inventory is

Conjectures 1, 2, 3, 4, 6, Questions 5, 7, 8, 9.

The results below prove every one of these statements. The unnumbered strong form (3) of Conjecture 2 is also proved.

2 Avoided first-relay quantities

Fix an avoided set $Y \subseteq V$. For a vertex x put

$$D_Y(x) := \{x \not\leftrightarrow Y\}, \quad d_Y(x) := \mathbb{P}(D_Y(x)).$$

For a source set S , define analogously

$$D_Y(S) := \{S \not\leftrightarrow Y\} = \bigcap_{s \in S} D_Y(s).$$

Let $T \subseteq V \setminus Y$ be a relay set and let $r : T \rightarrow \{1, \dots, |T|\}$ be a bijective rank. For an increasing functional Φ , assume first that $d_Y(x) > 0$ for every $x \in T$ and put

$$m_x^Y := \frac{\mathbb{E}[\Phi(C_x); D_Y(x)]}{d_Y(x)}. \quad (8)$$

The rank is called compatible if

$$r(x) < r(z) \implies m_x^Y \leq m_z^Y. \quad (9)$$

The first-relay pattern of x , seen from S while Y is avoided, is

$$J_x^{S,Y} := D_Y(S) \cap \{S \leftrightarrow x\} \cap \bigcap_{\substack{z \in T \\ r(z) < r(x)}} \{S \not\leftrightarrow z\}. \quad (10)$$

The patterns are disjoint and

$$\bigsqcup_{x \in T} J_x^{S,Y} = D_Y(S) \cap \{S \leftrightarrow T\}. \quad (11)$$

Define the avoided source surplus

$$\text{Sur}_{S,Y}(T; r, \Phi) := \mathbb{E}[\Phi(C_S); D_Y(S), S \leftrightarrow T] - \sum_{x \in T} \mathbb{P}(J_x^{S,Y}) m_x^Y. \quad (12)$$

For a singleton source we abbreviate $\text{Sur}_{x,Y} := \text{Sur}_{\{x\},Y}$. Notice that no connectedness of C_S is asserted or needed.

3 The source-set lift of the conditioned slack hierarchy

The extension needed below is not obtained by wiring the source vertices. Wiring would turn C_S into one connected cluster, but would also change the law of a relay cluster whenever that relay reaches S . There is a second, subtler trap. From

$$z \not\leftrightarrow B, \quad z \leftrightarrow S$$

one cannot conclude $S \not\leftrightarrow B$: a different component generated by S may hit B . Thus an unguarded set-observer version of CSH is not enough. The correct extension retains the full source-avoidance guard at every evaluation.

3.1 Guarded source forms

For finite vertex sets R, X, Y , write

$$\mathbf{H}_{X|Y}(R) := \{R \not\leftrightarrow Y, R \leftrightarrow X\}. \quad (13)$$

The elementary identity driving the construction is the disjoint union

$$\mathbf{H}_{X \cup \{d\}|Y}(R) = \mathbf{H}_{X|Y}(R) \sqcup \mathbf{H}_{\{d\}|Y \cup X}(R). \quad (14)$$

Indeed, if C_R has not met X , then its first new contact with $X \cup \{d\}$ is a contact with d , and in that case C_R avoids $Y \cup X$. Notice that (14) keeps track of every component in C_R .

For a vertex owner x , an avoided set Y , a finite source R , and an increasing functional ψ , put

$$d_x(Y) := \mathbb{P}(x \not\leftrightarrow Y), \quad (15)$$

$$\begin{aligned} \Gamma_{x,Y}^\psi(R) &:= d_x(Y) \mathbb{E}[\psi(C_x); \mathbf{H}_{\{x\}|Y}(R)] \\ &\quad - \mathbb{E}[\psi(C_x); x \not\leftrightarrow Y] \mathbb{P}(\mathbf{H}_{\{x\}|Y}(R)). \end{aligned} \quad (16)$$

When $R = \{u\}$, the guard and the contact imply that $C_u = C_x$ and $x \not\leftrightarrow Y$; hence (16) is exactly the ordinary denominator-free CSH covariance evaluated at u .

Let $\mathbf{d} = (d_1, \dots, d_\ell)$ be a list of distinct decoys and set

$$B_j = Y \cup \{x, d_i : i < j\}, \quad B_* = Y \cup \{x, d_1, \dots, d_\ell\}.$$

The guarded decoy profile and observer price are

$$c_j(R) = \frac{\mathbb{P}(\mathbf{H}_{\{d_j\}|B_j}(R))}{\mathbb{P}(d_j \not\leftrightarrow B_j)}, \quad (17)$$

$$p_{O,v} = \frac{\mathbb{P}(\mathbf{H}_{\{v\}|B_*}(O))}{\mathbb{P}(v \not\leftrightarrow B_*)}. \quad (18)$$

For a function h on finite source sets define

$$L_0 h = h, \quad (L_j h)(R) = (L_{j-1} h)(R) - c_j(R) (L_{j-1} h)(\{d_j\}). \quad (19)$$

All evaluations other than the primary one at O are singletons. In particular, for singleton $O = \{o\}$ all guards in (17)–(18) collapse by connectivity, and these are precisely the constants of the CSH already in the project.

Theorem 3.1 (guarded-source conditioned slack hierarchy). *Assume first that all edge weights lie in $(0, 1)$ and that the sets Y , O , $\{v\}$, $\{x\}$, and $\{d_1, \dots, d_\ell\}$ are pairwise disjoint (the d_j are distinct). For every increasing ψ ,*

$$\boxed{L_\ell \Gamma_{x,Y}^\psi(O) - p_{O,v} L_\ell \Gamma_{x,Y}^\psi(\{v\}) \geq 0.} \quad (20)$$

Proof. We give every additional identity needed beyond the existing CSH proof. Throughout this proof a covariance is unnormalised: thus $\text{cov}_D(f, g) = \mathbb{P}(D) \mathbb{E}[fg; D] - \mathbb{E}[f; D] \mathbb{E}[g; D]$. Every Γ in the asserted margin annihilates constants. Since the state space is finite, we may therefore add a constant to ψ and assume throughout the proof that $\psi \geq 0$.

1. *The source kernel and guarded exchange.* Work in an induced finite world U . For a possible cluster K of a vertex v , define

$$\alpha_{R,B}^U(K) = \mathbf{1}\{K \cap R \neq \emptyset\} \mathbb{P}_{U \setminus K}(C_{R \setminus K} \cap B = \emptyset). \quad (21)$$

Boundary pairs meeting K are deleted in $U \setminus K$. Directly from this definition,

$$K \subseteq K' \implies \alpha_{R,B}^U(K) \leq \alpha_{R,B}^U(K'), \quad B \subseteq B' \implies \alpha_{R,B'}^U(K) \leq \alpha_{R,B}^U(K). \quad (22)$$

For the first implication, the indicator can only turn on, the residual source shrinks, and the residual graph loses possible paths to B . Exploration of C_v gives the exact tower identity

$$\mathbb{E}_U[\alpha_{R,B}^U(C_v); v \not\leftrightarrow B] = \mathbb{P}_U(R \not\leftrightarrow B, R \leftrightarrow v). \quad (23)$$

In particular, exploration of C_x gives

$$\Gamma_{x,Y}^\psi(R) = d_x(Y) \mathbb{E}[\psi(C_x) \alpha_{R,Y}(C_x); x \not\leftrightarrow Y] - \mathbb{E}[\psi(C_x); x \not\leftrightarrow Y] \mathbb{E}[\alpha_{R,Y}(C_x); x \not\leftrightarrow Y]. \quad (24)$$

Thus a guarded contact can be averaged to an increasing function of one cluster. We use this identity below for the exchange estimate. It is important, however, not to feed $\alpha_{R,Y}^V$ unchanged into the old one-cluster Gibbs reduction: after C_Y is exposed, that fixed kernel is not the residual-world contact indicator. The correct reduction is the two-cluster calculation in part 4 below.

Fix a marker set X with $x \in X$ and $v \notin X$, and put

$$E_{X,R}(N) = \mathbb{P}_U(R \not\leftrightarrow X \cup N, R \leftrightarrow v), \quad M_X(N) = \mathbb{P}_U(v \not\leftrightarrow X \cup N).$$

If $N \cap N' = \emptyset$ and N, N' are disjoint from X , then

$$E_{X,R}(N) M_X(N') \leq M_X(N \cup N') E_{X,R}(\emptyset). \quad (25)$$

Here is a complete derivation. The one-cluster BHK core says, for increasing nonnegative f, g ,

$$\mathbb{E}[f(C_v); v \not\leftrightarrow A] \mathbb{E}[g(C_v); v \not\leftrightarrow A'] \leq \mathbb{E}[f(C_v)g(C_v); v \not\leftrightarrow A \cap A'] \mathbb{P}(v \not\leftrightarrow A \cup A').$$

Apply it with avoided sets $X \cup N$ and $X \cup N'$, increasing function $K \mapsto \alpha_{R, X \cup N}^U(K)$, and the constant function 1. Because the two additional sets are disjoint, their intersection is X and their union is $X \cup N \cup N'$. Thus

$$\mathbb{E}[\alpha_{R, X \cup N}(C_v); v \not\leftrightarrow X \cup N] M_X(N') \leq \mathbb{E}[\alpha_{R, X \cup N}(C_v); v \not\leftrightarrow X] M_X(N \cup N').$$

The first expectation is $E_{X,R}(N)$ by (23). By (22), the second is at most the same expectation with $X \cup N$ replaced by X , namely $E_{X,R}(\emptyset)$. This proves (25). Cases in which the displayed disjointness fails reduce either to the same statement after deleting redundant vertices or to a zero factor.

2. *The guarded two-source inequality.* For a nonnegative world functional h write, with $U_N = U \setminus C_N$,

$$\begin{aligned} Y_h^x(N) &= \mathbb{E}_U[\mathbf{1}\{x \not\leftrightarrow N\} h(U_N)], \\ X_{R,h}^x(N) &= \mathbb{E}_U[\mathbf{1}\{x \not\leftrightarrow N, R \not\leftrightarrow N\} q_R(U_N) h(U_N)], \\ q_R(W) &= \frac{\mathbb{P}_W(R \not\leftrightarrow X, R \leftrightarrow v)}{\mathbb{P}_W(v \not\leftrightarrow X)}. \end{aligned}$$

The one-source stopped-Harris estimate already proved in the project is

$$Y_h^x(N)M_X(\emptyset) \leq M_X(N)h(U), \quad N \mapsto Y_h^x(N) \text{ is antitone,} \quad (26)$$

for $h(W) = \text{Cov}_W(\psi(C_x), \mathbf{1}\{v \leftrightarrow X\})$. The guarded analogue of `CovTau.metaA2_of_star` is

$$E_{X,R}(N)Y_h^x(N') \leq M_X(N \cup N')X_{R,h}^x(N \cap N'). \quad (27)$$

For $N \cap N' = \emptyset$, multiply (26) by $E_{X,R}(N)$ and use (25). Since $E_{X,R}(\emptyset) = q_R(U)M_X(\emptyset)$, cancellation of the positive factor $M_X(\emptyset)$ gives

$$E_{X,R}(N)Y_h^x(N') \leq M_X(N \cup N')q_R(U)h(U),$$

which is (27) at the empty intersection. If that factor is zero, both $E_{X,R}(N)$ and the left side vanish, so the same conclusion holds without division.

For completeness we give the nonempty-intersection step rather than referring to the analogous step in the ordinary META-A2 proof. Put $Z = N \cap N' \neq \emptyset$ and $U' = U \setminus Z$. The cases $R \cap Z \neq \emptyset$, $x \in Z$, or $v \in Z$ have a zero factor and are immediate. Otherwise expose the set $s \subseteq U'$ of vertices having at least one open pair to Z . The indicators defining s use disjoint blocks of pair coordinates, so its law is a product law. With $N_0 = N \setminus Z$ and $N'_0 = N' \setminus Z$, source-cluster exploration gives the following four exact residual functions:

$$\begin{aligned} e(s) &= E_{X,R}^{U'}(N_0 \cup s), & y(s) &= Y_h^{x,U'}(N'_0 \cup s), \\ m(s) &= M_X^{U'}(N_0 \cup N'_0 \cup s), & \chi(s) &= X_{R,h}^{x,U'}(s). \end{aligned}$$

The extra source guard has an exact recursion here: on the nonzero branch, $R \not\leftrightarrow N$ in U is equivalent to $R \not\leftrightarrow (N_0 \cup s)$ in U' . Thus no residual avoidance event has been factored.

For two possible neighbour sets $s, t \subseteq U'$, define the cross-deleted arguments

$$P = s \cup (N_0 \setminus t), \quad P' = t \cup (N'_0 \setminus s).$$

Both $E_{X,R}$ and Y_h^x are antitone. Since $P \subseteq N_0 \cup s$ and $P' \subseteq N'_0 \cup t$, nonnegativity and the induction hypothesis in U' give

$$e(s)y(t) \leq E_{X,R}^{U'}(P)Y_h^{x,U'}(P') \leq M_X^{U'}(P \cup P')X_{R,h}^{x,U'}(P \cap P').$$

As $N_0 \cap N'_0 = \emptyset$, elementary set algebra gives

$$P \cup P' = N_0 \cup N'_0 \cup (s \cup t), \quad P \cap P' = s \cap t.$$

Consequently

$$e(s)y(t) \leq m(s \cup t)\chi(s \cap t).$$

All four functions are nonnegative, and the product density of the neighbour set satisfies $\pi(s)\pi(t) = \pi(s \cap t)\pi(s \cup t)$. The Ahlswede–Daykin four-functions theorem therefore integrates this pointwise inequality to (27). Together with the empty intersection case, this proves (27) by strong induction on $|U|$.

Take $N = N' = Y$ in (27). With $U_Y = V \setminus C_Y$ it says

$$\begin{aligned} \mathbb{P}(R \not\leftrightarrow X \cup Y, R \leftrightarrow v) \mathbb{E}[\mathbf{1}\{x \not\leftrightarrow Y\}h(U_Y)] \\ \leq \mathbb{P}(v \not\leftrightarrow X \cup Y) \mathbb{E}[\mathbf{1}\{x \not\leftrightarrow Y, R \not\leftrightarrow Y\}q_R(U_Y)h(U_Y)]. \end{aligned} \quad (28)$$

We also need the following guarded version of the four-point transfer. Inside any fixed world put

$$h_R(U) = \text{Cov}_U(\psi(C_x), \mathbf{1}\{R \leftrightarrow X\}), \quad h_v(U) = \text{Cov}_U(\psi(C_x), \mathbf{1}\{v \leftrightarrow X\}).$$

Then

$$h_R(U) \geq q_R(U)h_v(U). \quad (29)$$

Here is a proof, included because the superficially stronger coefficient $\mathbb{P}(R \leftrightarrow v \mid v \not\leftrightarrow X)$ is in general false for a disconnected source R . Put

$$A = \{R \leftrightarrow X\}, \quad B = \{R \leftrightarrow v\}, \quad D = \{v \not\leftrightarrow X\}, \quad J = \{R \not\leftrightarrow X, R \leftrightarrow v\}.$$

The increasing event $A \cup B$ and Harris give, with $m = \mathbb{E}_U \psi(C_x)$,

$$h_R(U) \geq m\mathbb{P}_U(J) - \mathbb{E}_U[\psi(C_x); J], \quad (30)$$

because $(A \cup B) \setminus A = J$. On D the event J is

$$\{C_v \cap R \neq \emptyset\} \cap \{C_X \cap R = \emptyset\}.$$

It is increasing in C_v and decreasing in C_X , whereas $\psi(C_x)$ is constant in the first cluster and increasing in the second. The two-set BHK conditional association theorem therefore gives

$$\mathbb{P}_U(D)\mathbb{E}_U[\psi(C_x); J] \leq \mathbb{P}_U(J)\mathbb{E}_U[\psi(C_x); D]. \quad (31)$$

Since $h_v(U) = \mathbb{P}_U(D)m - \mathbb{E}_U[\psi(C_x); D]$, equations (30)–(31) prove (29); the zero-denominator case is again immediate.

Divide (28) by its positive denominator and combine it with (29). The resulting horizontal margin is

$$\mathbb{E}[\mathbf{1}\{x \not\leftrightarrow Y, R \not\leftrightarrow Y\}h_R(U_Y)] - p_{R,v}\mathbb{E}[\mathbf{1}\{x \not\leftrightarrow Y\}h_v(U_Y)] \geq 0. \quad (32)$$

3. The repaired guarded unfolding. We spell out the point at which the guard changes the existing `CSH.s1Form_jn` calculation. Fix an outer configuration, let Z be the already exposed cluster of Y , and work in the residual graph. For its connectivity relation write

$$j_{S|Z}(R) = \mathbf{1}\{R \cap Z = \emptyset, R \leftrightarrow S\}, \quad a_{S,d|Z}(R) = \mathbf{1}\{R \cap Z = \emptyset, R \not\leftrightarrow S, R \leftrightarrow d\}.$$

Then (14) is the pointwise identity

$$j_{S \cup \{d\}|Z}(R) = j_{S|Z}(R) + a_{S,d|Z}(R). \quad (33)$$

If $e_{d,Z} = \mathbf{1}\{d \notin Z\}$, the singleton dichotomy is

$$j_{S|Z}(\{d\}) = e_{d,Z} - a_{S,d|Z}(\{d\}). \quad (34)$$

Consequently one level step, with any coefficient function c , is

$$\begin{aligned} j_{S|Z}(R) - c(R)j_{S|Z}(\{d\}) \\ = j_{S \cup \{d\}|Z}(R) - (a_{S,d|Z}(R) - c(R)a_{S,d|Z}(\{d\})) - c(R)e_{d,Z}. \end{aligned} \quad (35)$$

The last term is constant in the residual configuration and therefore has zero covariance. Iterating (35) is the exact guarded replacement for `CSH.s1Form_jn`; it produces the marker $S = \{x, d_1, \dots, d_\ell\}$ and one centred bracket for each decoy.

We now record the extra exploration which is essential for a set-valued source. It is useful to distinguish edge and vertex clusters. Write $\mathcal{C}_R(\omega)$ for the union of the open *edge* clusters rooted in R , and put

$$V_R(F) := R \cup \{u : \exists e \in F, u \in e\}, \quad C_R(\omega) = V_R(\mathcal{C}_R(\omega)).$$

Thus isolated source vertices are retained. In the Lean notation let g be the edge-cluster reading of ψ and set

$$\Theta(\omega) := \text{CSH.resid}(w, x, Y, g, \omega), \quad \widehat{\Phi}(K) := \text{CSH.phiFun}(w, x, Y, g, K).$$

The correctly typed functional on an edge cluster of d is

$$\Psi_d(F) := \widehat{\Phi}(\{d\} \cup \{u : \exists e \in F, u \in e\}), \quad \Psi_d(\mathcal{C}_d) = \widehat{\Phi}(C_d). \quad (36)$$

The existing lemmas `CSH.phiFun_nonneg` and `CSH.phiFun_mono` say that $\widehat{\Phi}$, and hence Ψ_d , is nonnegative and increasing.

Let S contain x , let $d \notin S \cup Y$, and put $B = S \cup Y$. For an arbitrary finite source R abbreviate

$$E_R := \mathbf{H}_{\{d\}|B}(R) = \{C_R \cap B = \emptyset, d \in C_R\}, \quad E_d := \{d \not\leftrightarrow B\}.$$

The correct set-cluster Markov identity is

$$\mathbb{E}[\Theta; E_R] = -\mathbb{E}[\widehat{\Phi}(C_R); E_R]. \quad (37)$$

Here is a finite-sum proof, included to make the replacement formalizable. For an edge set F let

$$\iota(F) = \mathbf{1}\{V_R(F) \cap B = \emptyset, d \in V_R(F)\},$$

and let $\overline{F}_R$ be the set of pairs meeting $V_R(F)$. Part (a) of Lemma Φ , in residual form, gives

$$\sum_{\eta} \text{wt}(\eta) \Theta(\eta \setminus \overline{F}_R) = -\widehat{\Phi}(V_R(F)).$$

Apply the set-cluster conditioning identity `CSH.set_sum_cond_sdiff_off` to the kernel $(F, \beta) \mapsto \iota(F)\Theta(\beta)$. On $\{\iota(C_R) = 1\}$ the whole union C_R avoids $\{x\} \cup Y \subseteq B$, so deleting all pairs meeting C_R changes neither C_x , C_Y , nor the residual Θ . The left side of the conditioning identity is therefore the left side of (37); the displayed residual identity turns its right side into the right side of (37). The point is that the exploration is of the *whole* source union C_R , not only the one component C_d .

On E_R one has $C_d \subseteq C_R$. Define the unnormalised correction

$$\text{Err}_{d,B}(R) := \mathbb{E}[\widehat{\Phi}(C_R) - \Psi_d(\mathcal{C}_d); E_R]. \quad (38)$$

Monotonicity of $\widehat{\Phi}$ gives

$$\text{Err}_{d,B}(R) \geq 0. \quad (39)$$

For every vertex u ,

$$\text{Err}_{d,B}(\{u\}) = 0, \quad (40)$$

because on $E_{\{u\}}$ the connection $u \leftrightarrow d$ gives $C_u = C_d$. Impossible events and the case $u = d$ are included.

Equations (37)–(38) give

$$\mathbb{E}[\Theta; E_R] = -\mathbb{E}[\Psi_d(\mathcal{C}_d); E_R] - \text{Err}_{d,B}(R),$$

whereas the singleton source gives the exact identity

$$\mathbb{E}[\Theta; E_d] = -\mathbb{E}[\Psi_d(\mathcal{C}_d); E_d]. \quad (41)$$

Put $e_d = \mathbb{P}(E_d)$ and $q_R = \mathbb{P}(E_R)$. The division-free form, valid also when $e_d = 0$, is

$$e_d \mathbb{E}[\Theta; E_R] - q_R \mathbb{E}[\Theta; E_d] = -\Gamma_{d,B}^{\hat{\Phi}}(R) - e_d \text{Err}_{d,B}(R). \quad (42)$$

Under the present nondegeneracy assumptions $e_d > 0$. With $c(R) = q_R/e_d$, and after the C_Y Markov merge, the two a -terms are the indicators of E_R and E_d . Hence the corrected one-decoy identity is

$$\mathbb{E}[\Theta(a_{S,d|C_Y}(R) - c(R)a_{S,d|C_Y}(\{d\}))] = -e_d^{-1} \Gamma_{d,B}^{\hat{\Phi}}(R) - \text{Err}_{d,B}(R). \quad (43)$$

We now make every quantity in the unfolding explicit. Let μ be the original product law, put

$$D = \{x \not\rightarrow Y\}, \quad \delta = \mathbb{P}(D) > 0,$$

and, for an outer configuration $\omega \in D$, let $Z_\omega = C_Y(\omega)$. Denote by μ_ω the product law obtained from μ by setting to zero every pair meeting Z_ω . Thus μ_ω is the residual-world law after the Y -cluster has been exposed. For a residual configuration η define

$$\begin{aligned} j_S^\omega(R; \eta) &= \mathbf{1}\{R \cap Z_\omega = \emptyset, C_R(\eta) \cap S \neq \emptyset\}, \\ a_{S,d}^\omega(R; \eta) &= \mathbf{1}\{R \cap Z_\omega = \emptyset, C_R(\eta) \cap S = \emptyset, d \in C_R(\eta)\}, \\ \kappa_S^\omega(R) &= \text{Cov}_{\mu_\omega}(\psi(C_x), j_S^\omega(R; \cdot)). \end{aligned}$$

The cluster symbols in these three formulas are evaluated in the residual configuration. These are precisely $j_{S|Z_\omega}$, $a_{S,d|Z_\omega}$, and their world covariance from (35).

We shall also need tail level forms. For $0 \leq j \leq k \leq \ell$ put

$$L_{j,j}h = h, \quad (L_{j,k}h)(R) = (L_{j,k-1}h)(R) - c_k(R)(L_{j,k-1}h)(\{d_k\}), \quad L_{>j} := L_{j,\ell}.$$

Thus $L_{>0} = L_\ell$, while $L_{>j}$ uses exactly the later decoys $d_{j+1}, \dots, d_\ell$. Write

$$X = \{x, d_1, \dots, d_\ell\}, \quad p = p_{O,v}.$$

The unnormalised averaged within-world and horizontal margins are

$$\mathcal{W}_{x,Y,d}^{O,v}(\psi) := \int_D \left[(L_{>0} \kappa_{\{x\}}^\omega)(O) - p(L_{>0} \kappa_{\{x\}}^\omega)(\{v\}) \right] d\mu(\omega), \quad (44)$$

$$\mathcal{H}_{x,Y,d}^{O,v}(\psi) := \int_D \left[\kappa_X^\omega(O) - p\kappa_X^\omega(\{v\}) \right] d\mu(\omega). \quad (45)$$

By cluster exploration, the second quantity is exactly

$$\mathbb{E}[\mathbf{1}\{x \not\rightarrow Y, O \not\rightarrow Y\} h_O(U_Y)] - p\mathbb{E}[\mathbf{1}\{x \not\rightarrow Y\} h_v(U_Y)],$$

the left side of (32) with $R = O$. If $v \in C_Y$, its residual-world contact indicator is zero. Hence $\mathcal{H} \geq 0$.

For the j -th decoy set

$$S_j = \{x, d_i : i < j\}, \quad B_j = S_j \cup Y, \quad e_j = \mathbb{P}(d_j \not\rightarrow B_j), \quad \text{Err}_j = \text{Err}_{d_j, B_j}, \quad \Psi_j = \Psi_{d_j}.$$

The exact bridge between the residual-world calculation and the global one-decoy identity is the following function of the source R :

$$\begin{aligned}
I_j(R) &:= \int_D \text{Cov}_{\mu_\omega} \left(\psi(C_x), a_{S_j, d_j}^\omega(R; \cdot) - c_j(R) a_{S_j, d_j}^\omega(\{d_j\}; \cdot) \right) d\mu(\omega) \\
&= \mathbb{E}[\Theta; \mathbf{H}_{\{d_j\}|B_j}(R)] - c_j(R) \mathbb{E}[\Theta; d_j \not\rightarrow B_j] \\
&= -e_j^{-1} \Gamma_{d_j, B_j}^{\widehat{\Phi}}(R) - \text{Err}_j(R).
\end{aligned} \tag{46}$$

The second line is a kernel-valued C_Y set-cluster Markov merge; unlike a fixed-test conditioning statement, it retains the full guard $R \not\rightarrow Y$. For completeness, here is the finite-sum identity being used. Write $\text{cut}(Z)$ for all pairs meeting Z and put

$$\bar{g}(Z) = \sum_{\eta} \text{wt}(\eta) \psi(C_x(\eta \setminus \text{cut}(Z))).$$

For every bounded two-argument test F set-cluster conditioning gives

$$\begin{aligned}
\sum_{\omega} \text{wt}(\omega) \mathbf{1}\{x \notin Z_\omega\} \sum_{\eta} \text{wt}(\eta) [\psi(C_x(\eta \setminus \text{cut}(Z_\omega))) - \bar{g}(Z_\omega)] F(Z_\omega, \eta \setminus \text{cut}(Z_\omega)) \\
= \sum_{\zeta} \text{wt}(\zeta) \Theta(\zeta) F(Z_\zeta, \zeta \setminus \text{cut}(Z_\zeta)).
\end{aligned} \tag{47}$$

This is the set-cluster tower identity applied to the kernel $(Z, \beta) \mapsto \mathbf{1}\{x \notin Z\}(\psi(C_x(\beta)) - \bar{g}(Z))F(Z, \beta)$. On the guarded branch, $a_{S_j, d_j}^\zeta(R; \zeta \setminus \text{cut}(Z_\zeta))$ is exactly the indicator of $\mathbf{H}_{\{d_j\}|S_j \cup Y}(R)$ in ζ . This proves the second line of (46); its third line is (43).

Define the lower-level margin

$$\mathcal{M}_j := (L_{>j} \Gamma_{d_j, B_j}^{\widehat{\Phi}})(O) - p(L_{>j} \Gamma_{d_j, B_j}^{\widehat{\Phi}})(\{v\}). \tag{48}$$

There is no change of constants hidden here: for every $k > j$ the tail profile in the guarded hierarchy with owner d_j , avoided set B_j , and tail $(d_{j+1}, \dots, d_\ell)$ is c_k , and

$$B_j \cup \{d_j, d_{j+1}, \dots, d_\ell\} = B_*.$$

Consequently its observer price is the same p , and $\mathcal{M}_j$ is exactly the left side of (20) for that lower-level datum.

Iterating the pointwise identity (35), taking world covariances, and then performing the final $O - pv$ subtraction gives

$$\mathcal{W} = \mathcal{H} - \sum_{j=1}^{\ell} \left[(L_{>j} I_j)(O) - p(L_{>j} I_j)(\{v\}) \right]. \tag{49}$$

All omitted terms here are constant in the residual configuration and therefore have zero world covariance. Moreover (40) implies, by induction through the tail steps, that $L_{>j} \text{Err}_j = \text{Err}_j$. Substitution of (46) into (49) therefore gives the exact repaired unfolding

$$\mathcal{W}_{x, Y, \mathbf{d}}^{O, v}(\psi) = \mathcal{H}_{x, Y, \mathbf{d}}^{O, v}(\psi) + \sum_{j=1}^{\ell} e_j^{-1} \mathcal{M}_j + \sum_{j=1}^{\ell} \text{Err}_j(O). \tag{50}$$

In particular,

$$\mathcal{W}_{x,Y,d}^{O,v}(\psi) \geq \mathcal{H}_{x,Y,d}^{O,v}(\psi) + \sum_{j=1}^{\ell} e_j^{-1} \mathcal{M}_j. \quad (51)$$

We prove Theorem 3.1 by strong induction on ℓ . If $\ell = 0$, then $\mathcal{W} = \mathcal{H} \geq 0$, and part 4 below turns this within-world inequality into the level-zero theorem. For $\ell > 0$, assume the theorem for every shorter decoy list. Each $\mathcal{M}_j$ has only $\ell - j$ decoys and is nonnegative by that induction hypothesis, applied to the increasing vertex functional $\widehat{\Phi}$; its correctly typed edge-cluster reading in Lean is Ψ_j . Since $e_j > 0$ and every error is nonnegative, (50) gives $\mathcal{W} \geq 0$. Part 4 turns this into the level- ℓ guarded CSH margin and closes the induction.

Remark 3.2 (regression test for the repaired identity). The error term cannot be omitted, even with all weights strictly between zero and one. Take $V = \{x, d, r_1, r_2\}$, give the pairs $\{x, r_2\}$ and $\{d, r_1\}$ weight $1/2$ and every other pair weight $1/4$, and let $Y = \emptyset$, $S = \{x\}$, $R = \{r_1, r_2\}$, and $g(C_x) = \mathbf{1}\{r_2 \in C_x\}$. Exact enumeration gives

$$\mathbb{E}[\Theta_{a_{S,d}}(R)] = -\frac{1665}{16384}, \quad -\mathbb{E}[\widehat{\Phi}(C_d); E_R] = -\frac{1017}{16384},$$

and their discrepancy is $-81/2048 = -\text{Err}_{d,B}(R)$. Thus the former C_d -only equality is false, while (37)–(43) give the exact replacement.

4. *The two-cluster reduction, including the missing half-step.* This is the point at which the source guard must be retained. Recall $D = \{x \not\sim Y\}$ and $\delta = \mathbb{P}(D) > 0$. Under the conditional law $\mathbb{P}(\cdot \mid D)$ put $A = C_x$ and $B = C_Y$, viewed as their vertex sets, and define

$$b_R(A, B) = \mathbf{1}\{A \cap R \neq \emptyset\} \mathbf{1}\{B \cap R = \emptyset\}.$$

Then $b_R(C_x, C_Y)$ is exactly the indicator of $\mathbf{H}_{\{x\}|Y}(R)$. Apply the level form and the final observer subtraction to $R \mapsto b_R(A, B)$, and call the resulting bounded test $\mathfrak{h}(A, B)$. Its coefficient at the primary source O is $+1$; every other evaluation is at a singleton $u \in \{d_1, \dots, d_\ell, v\}$. Since $A \cap B = \emptyset$ on D ,

$$\mathfrak{h}(A, B) = b_O(A, B) + \sum_{u \in \{d_1, \dots, d_\ell, v\}} \lambda_u \mathbf{1}\{u \in A\} \quad (52)$$

for constants λ_u . In particular, for fixed A the map $B \mapsto \mathfrak{h}(A, B)$ is antitone. This observation is why the argument works for one set-valued primary source but keeps all decoys and the second observer singleton-valued.

Here is the exact reduction lemma. Under the conditional law on D , let P_A and P_B denote conditional expectation given A and B , respectively, and put $T = P_A P_B$ on functions of A . For every bounded $g = g(A)$, with $u = P_B g$, the elementary conditional-covariance identity is

$$\begin{aligned} \text{Cov}_D(g, \mathfrak{h}) - \text{Cov}_D(Tg, \mathfrak{h}) \\ = \mathbb{E}_D[\text{Cov}(g, \mathfrak{h} \mid B)] + \mathbb{E}_D[\text{Cov}(u, \mathfrak{h} \mid A)]. \end{aligned} \quad (53)$$

The conditional cluster decomposition and (44) give the exact normalization

$$\mathbb{E}_D[\text{Cov}(g, \mathfrak{h} \mid B)] = \delta^{-1} \mathcal{W}_{x,Y,d}^{O,v}(g).$$

Here and below $\mathcal{W}(g)$ denotes (44) with the cluster functional ψ replaced by g . Thus the first term is nonnegative. The factor δ^{-1} is essential: $\mathcal{W}$ is the unnormalised integral over D , whereas (53) is written under the conditional law. If g is increasing, then $u(B) = \mathbb{E}[g(A) \mid B]$ is antitone:

conditionally on B , the cluster A is the x -cluster in the graph with every pair meeting B deleted, and the common-edge coupling for $B \subseteq B'$ makes the latter cluster smaller. This is the existing lemma `BHK2006.condS_antitone`. By (52), $B \mapsto \mathfrak{h}(A, B)$ is also antitone for fixed A . Conditional on A , write $B = \text{BHK2006.halfT}(A, \eta)$, where η is a fresh product configuration in the graph with the pairs meeting A deleted. The map $\eta \mapsto B$ is increasing. Thus both $u(B)$ and $\mathfrak{h}(A, B)$ are decreasing functions of η , and `BHK2006.harris_anti_anti` makes the second term in (53) nonnegative.

For clarity, the other property needed in the iteration follows from the same coupling. If $A \subseteq A'$, then the conditional Y -cluster sampled after deleting A' is contained in the one sampled after deleting A . Since u is antitone,

$$(Tg)(A) = \mathbb{E}[u(B) \mid A] \leq \mathbb{E}[u(B) \mid A'] = (Tg)(A').$$

Hence the BHK Gibbs operator T preserves monotonicity; it plainly preserves constants and bounds. Under the present nondegeneracy assumption it also has a uniform regeneration. If $Y \neq \emptyset$, close every pair meeting Y in the B -resampling half-step. This has conditional probability at least

$$\epsilon := \prod_{e: e \cap Y \neq \emptyset} (1 - p_e) > 0,$$

independently of the old A , and forces $B = Y$. The following A -resampling then has a fixed law, independent of the old state. Thus the transition kernel is of the form $\epsilon\nu + (1 - \epsilon)Q_A$, and

$$\text{osc}(T^n g) \leq (1 - \epsilon)^n \text{osc}(g).$$

If $Y = \emptyset$, the first application of T is already constant. Since minima of $T^n g$ increase and maxima decrease, $T^n g$ converges uniformly to a constant. The same statements apply with g replaced by $T^n g$: the operator T preserves nonnegativity and monotonicity, so the first term in (53) is $\delta^{-1}\mathcal{W}(T^n g) \geq 0$, and the second is nonnegative by the same antitone–antitone Harris argument. Iterating (53) consequently gives

$$\text{Cov}_D(g, \mathfrak{h}) \geq \text{Cov}_D(T^n g, \mathfrak{h}) \longrightarrow 0.$$

Finally, direct expansion of (16) and linearity of the level forms give

$$L_\ell \Gamma_{x,Y}^\psi(O) - p L_\ell \Gamma_{x,Y}^\psi(\{v\}) = \delta^2 \text{Cov}_D(g, \mathfrak{h}).$$

Thus the last nonnegative covariance is exactly the guarded CSH margin up to the positive factor δ^2 . This proves the theorem. Notice that this two-half-step calculation is essential: a fixed global kernel $\alpha_{R,Y}^V$ does *not* become $j_{\{x\}|Z}(R)$ merely by conditioning on $C_Y = Z$. $\square$

Remark 3.3 (what is new, and what is reused). The genuinely new declarations are the guarded event (13), the kernel (21), the exchange lemma (25), the additional $R \not\leftrightarrow N$ guard in the generic two-source induction, the set-source Markov identity (37), its nonnegative correction (38), and the two-half-step reduction (53). The BHK four-product theorem, set-cluster exploration, stopped Harris, Ahlswede–Daykin, the general Gibbs reduction, and the residual functional are already proved in the repository. In particular, Theorem 3.1 does not assume a new correlation conjecture. Omitting the guard would make the later peeling step false; factoring the unevaluated source components would incorrectly delete the positive error in (50).

3.2 Set-source surplus transfer

For $B \subseteq V$ write $D_B(k) = \{k \not\leftrightarrow B\}$. If $k \notin Y \cup T$, put

$$E_k := D_{Y \cup T}(k), \quad Q_{S,k} := D_{Y \cup T}(S) \cap \{S \leftrightarrow k\}. \quad (54)$$

For later use abbreviate the guarded one-decoy profile by

$$c_{z,B}(R) := \frac{\mathbb{P}(\mathbf{H}_{\{z\}|B}(R))}{\mathbb{P}(z \not\leftrightarrow B)}. \quad (55)$$

For a relay set T and an additional decoy list $\mathbf{d} = (d_1, \dots, d_\ell)$, set

$$B_j = Y \cup T \cup \{d_i : i < j\}, \quad B_* = Y \cup T \cup \{d_1, \dots, d_\ell\}.$$

Use the profiles c_{d_j, B_j} in the recursion (19) and the price

$$p_{O,v}^{Y,T,\mathbf{d}} = \frac{\mathbb{P}(\mathbf{H}_{\{v\}|B_*}(O))}{\mathbb{P}(v \not\leftrightarrow B_*)}.$$

We write

$$\text{Marg}_{Y,T,\mathbf{d}}^{O,v}[h] := L_\ell h(O) - p_{O,v}^{Y,T,\mathbf{d}} L_\ell h(\{v\}). \quad (56)$$

This definition removes a possible ambiguity below: the base avoided set for a surplus margin is $Y \cup T$, not merely Y .

Theorem 3.4 (set-source avoided surplus transfer). *Let Φ be increasing and let r be compatible with the conditional means (8). Suppose that $S, Y, T, \{k\}$ are pairwise disjoint. Then, for arbitrary edge weights in $[0, 1]$,*

$$\mathbb{P}(E_k) \text{Sur}_{S,Y}(T; r, \Phi) \geq \mathbb{P}(Q_{S,k}) \text{Sur}_{k,Y}(T; r, \Phi). \quad (57)$$

Proof. First take nondegenerate weights and decoys disjoint from all displayed sets. We prove, simultaneously for every such decoy list $\mathbf{d}$ and singleton v , the stronger statement

$$\text{Marg}_{Y,T,\mathbf{d}}^{S,v}[R \mapsto \text{Sur}_{R,Y}(T; r, \Phi)] \geq 0. \quad (58)$$

Induct on $|T|$.

If $T = \emptyset$, then $R \leftrightarrow T$ is impossible and $\text{Sur}_{R,Y}(T) = 0$ for every source R . Thus (58), and hence (57), holds with equality. Assume from now on that $T \neq \emptyset$.

Let z have maximal rank and write $T' = T \setminus \{z\}$. Set

$$E_z = D_{Y \cup T'}(z), \quad d_z = \mathbb{P}(E_z), \quad \kappa_z = m_z^Y d_z - \mathbb{E}[\Phi(C_z); E_z]$$

and

$$\Theta_z(R) = \mathbb{E}[(\Phi(C_z) - m_z^Y); \mathbf{H}_{\{z\}|Y \cup T'}(R)]. \quad (59)$$

For a singleton $R = \{u\}$, the usual first-pattern partition gives the exact peeling identity

$$\text{Sur}_{u,Y}(T) = \text{Sur}_{u,Y}(T') + \Theta_z(\{u\}). \quad (60)$$

For $R = S$, the same partition leaves $\Phi(C_S)$ in the new term, on the *same guarded event* appearing in (59). On that event $S \leftrightarrow z$, hence $C_z \subseteq C_S$; monotonicity gives the one-sided inequality

$$\text{Sur}_{S,Y}(T) \geq \text{Sur}_{S,Y}(T') + \Theta_z(S). \quad (61)$$

This is the only point at which a source union differs from a singleton. In the level margin the value at S occurs with coefficient $+1$ and all other evaluations are the singleton decoys and v . Therefore (61) and (60) preserve the inequality after applying the level form.

For every finite source R , direct expansion of (16) gives

$$d_z \Theta_z(R) = \Gamma_{z, Y \cup T'}^\Phi(R) - \kappa_z d_z c_{z, Y \cup T'}(R). \quad (62)$$

The guarded-source CSH theorem makes the margin of the first term nonnegative. Apply it once more to the increasing vertex-cluster functional

$$\psi_{\text{iso}, z}(K) = \mathbf{1}\{K \setminus \{z\} \neq \emptyset\}.$$

Every evaluated source is distinct from z . If $D = \{z \not\leftrightarrow Y \cup T'\}$ and $\rho_z = \mathbb{P}(D, C_z = \{z\})$, direct expansion gives

$$\Gamma_{z, Y \cup T'}^{\psi_{\text{iso}, z}}(R) = \rho_z \mathbb{P}(\mathbf{H}_{\{z\} | Y \cup T'}(R)) = \rho_z d_z c_{z, Y \cup T'}(R). \quad (63)$$

Under nondegenerate weights $\rho_z d_z > 0$. Guarded CSH for $\psi_{\text{iso}, z}$ therefore says precisely that the required margin of $c_{z, Y \cup T'}$ is nonnegative. Using a vertex-cluster indicator here of the event $K \neq \emptyset$ would be wrong, because a vertex cluster is never empty; the isolated state is $C_z = \{z\}$.

Finally, centering over $D_Y(z)$ and partitioning according as z hits T' or avoids it gives

$$\text{Sur}_{z, Y}(T') - \kappa_z = \sum_{x \in T'} \mathbb{P}(J_x^{\{z\}, Y})(m_z^Y - m_x^Y) \geq 0. \quad (64)$$

The last inequality is exactly maximality of the rank of z . Prepending z to the decoy list gives the algebraic identity

$$\text{Marg}_{z::\mathbf{d}}[\text{Sur}(T')] = \text{Marg}_{\mathbf{d}}[\text{Sur}(T')] - \text{Sur}_{z, Y}(T') \text{Marg}_{\mathbf{d}}[c_z].$$

Equations (61)–(64), the two nonnegative CSH margins, and the induction hypothesis therefore give

$$d_z \text{Marg}_{\mathbf{d}}[\text{Sur}(T)] \geq d_z \text{Marg}_{z::\mathbf{d}}[\text{Sur}(T')] \geq 0.$$

At decoy level zero, multiply out the observer price. With $v = k$ this is precisely (57).

It remains to remove the nondegeneracy restriction. This closure is proved explicitly just below; it applies to the cross-multiplied inequality (57) and completes the proof. $\square$

3.3 Ranks, zero denominators, and boundary weights

The closure used above is short but essential. Call $x \in T$ active if $d_Y(x) > 0$. If x is inactive, every event $D_Y(S) \cap \{S \leftrightarrow x\}$ is null, because it is contained in $D_Y(x)$. Inactive relays can therefore be deleted without changing any integral or first-pattern term. For the active relays, every compatible rank gives the rank-free identity

$$\text{Sur}_{S, Y}(T; r, \Phi) = \mathbb{E} \left[\mathbf{1}\{D_Y(S), S \leftrightarrow T\} \left(\Phi(C_S) - \min_{x \in T \cap C_S} m_x^Y \right) \right]. \quad (65)$$

Indeed, the first relay in C_S has the least conditional mean among $T \cap C_S$; ties have the same value. Formula (65) also proves independence from tie-breaking.

Approximate an arbitrary weight vector $w \in [0, 1]^E$ by nondegenerate vectors $w_n \in (0, 1)^E$. There are only finitely many relay orders, so pass to a subsequence on which the compatible order is fixed. For every relay active at w , its avoidance probability stays positive and its conditional

mean converges to the mean at w . Terms involving a relay inactive at w are supported on $D_Y(x)$ and hence tend to zero. Thus both surpluses in (57) converge to the active-relay expressions at w . The two probability prefactors are polynomials in the edge weights. Passing to the limit proves (57) at w . This argument neither chooses a limiting minimizer nor divides by a quantity that may vanish.

3.4 The conditional top-relay estimate

The following elementary lemma is useful both for checking the signs and for formalization.

Lemma 3.5 (set-source top-relay estimate). *Let $T_0 \subseteq V \setminus Y$, let $k \notin Y \cup T_0$, and set*

$$E = D_{Y \cup T_0}(k), \quad J = D_{Y \cup T_0}(S) \cap \{S \leftrightarrow k\}, \quad \kappa = m_k^Y \mathbb{P}(E) - \mathbb{E}[\Phi(C_k); E].$$

Then

$$\mathbb{P}(E) \mathbb{E}[(\Phi(C_S) - m_k^Y); J] \geq -\mathbb{P}(J)\kappa. \quad (66)$$

Proof. On J one has $C_k \subseteq C_S$, hence it is enough to replace $\Phi(C_S)$ by $\Phi(C_k)$ on the left.

We claim

$$\mathbb{P}(E) \mathbb{E}[\Phi(C_k); J] \geq \mathbb{P}(J) \mathbb{E}[\Phi(C_k); E]. \quad (67)$$

To see this directly, condition on E and then on $C_k = K$. Here $K \cap (Y \cup T_0) = \emptyset$. Given K , the unexplored edges off K are fresh, and the conditional probability of J is

$$q(K) = \mathbf{1}\{K \cap S \neq \emptyset\} \mathbb{P}_{G \setminus K}(C_{S \setminus K} \cap (Y \cup T_0) = \emptyset),$$

where $G \setminus K$ means that all pairs meeting K are deleted. If $K \subseteq K'$ and both avoid $Y \cup T_0$, then $q(K) \leq q(K')$: the first indicator can only increase, the residual source shrinks, and the residual graph loses edges. Thus both $K \mapsto \Phi(K)$ and $q(K)$ are increasing. Conditional positive association for C_k given $k \notin Y \cup T_0$ gives (67).

Subtracting $m_k^Y \mathbb{P}(E) \mathbb{P}(J)$ from the two sides of (67) gives (66). $\square$

3.5 Source-set avoided GEN

The transfer theorem was stated for disjoint sources and relays. The following one-cluster estimate handles all overlaps and is also useful as a separate formalization target.

Lemma 3.6 (overlapping source relay). *Let $t \in S \setminus Y$ and let Φ be increasing. Then*

$$d_Y(t) \mathbb{E}[\Phi(C_S); D_Y(S)] \geq \mathbb{P}(D_Y(S)) \mathbb{E}[\Phi(C_t); D_Y(t)]. \quad (68)$$

Proof. If $d_Y(t) = 0$, then $D_Y(S) \subseteq D_Y(t)$ is null and both sides vanish. Suppose it is positive. For a possible cluster $K = C_t$ with $K \cap Y = \emptyset$, put

$$\alpha(K) = \mathbb{P}_{G \setminus K}(C_{S \setminus K} \cap Y = \emptyset).$$

If $K \subseteq K'$, the residual source shrinks and the residual graph loses pairs, so $\alpha(K) \leq \alpha(K')$. Exploration of C_t gives the two exact tower identities

$$\begin{aligned} \mathbb{E}[\alpha(C_t); D_Y(t)] &= \mathbb{P}(D_Y(S)), \\ \mathbb{E}[\Phi(C_t)\alpha(C_t); D_Y(t)] &= \mathbb{E}[\Phi(C_t); D_Y(S)]. \end{aligned}$$

Conditional positive association of C_t given $D_Y(t)$, applied to the increasing functions $K \mapsto \Phi(K)$ and $K \mapsto \alpha(K)$, therefore yields

$$d_Y(t)\mathbb{E}[\Phi(C_t); D_Y(S)] \geq \mathbb{E}[\Phi(C_t); D_Y(t)]\mathbb{P}(D_Y(S)).$$

The association statement is unchanged after adding a constant to Φ , so no sign assumption is needed. Finally $C_t \subseteq C_S$ pointwise, and monotonicity of Φ proves (68). $\square$

Theorem 3.7 (source-set avoided first-relay inequality). *Let $S, Y, T \subseteq V$, with $T \cap Y = \emptyset$. Let $\Phi : 2^V \rightarrow \mathbb{R}$ be increasing, and rank T compatibly with the conditional means m_x^Y . Then*

$$\sum_{x \in T} \mathbb{P}(J_x^{S,Y}) m_x^Y \leq \mathbb{E}[\Phi(C_S); D_Y(S), S \leftrightarrow T]. \quad (69)$$

Equivalently, $\text{Sur}_{S,Y}(T; r, \Phi) \geq 0$.

Proof. If $S \cap Y \neq \emptyset$, then $D_Y(S)$ is empty and both sides of (69) vanish. Suppose next that $t \in S \cap T$. If t is inactive, then $D_Y(S) \subseteq D_Y(t)$ is null. Otherwise $S \leftrightarrow T$ is automatic, and on every first-relay pattern the relay met first has conditional mean at most m_t^Y , since $t \in C_S$ is itself available. Hence Lemma 3.6 gives

$$\sum_{x \in T} \mathbb{P}(J_x^{S,Y}) m_x^Y \leq \mathbb{P}(D_Y(S)) m_t^Y \leq \mathbb{E}[\Phi(C_S); D_Y(S)],$$

which is (69). We may therefore assume for the rest of the proof that S, Y, T are pairwise disjoint; this is exactly the domain in which Theorem 3.4 is used.

Adding a constant to Φ adds that constant times the two equal quantities in (11); hence we may assume $\Phi \geq 0$. The empty relay case is immediate. Suppose $T \neq \emptyset$.

Let k be the rank-maximal relay and put $T_0 = T \setminus \{k\}$. Set

$$E = D_{Y \cup T_0}(k), \quad J = D_{Y \cup T_0}(S) \cap \{S \leftrightarrow k\},$$

and define κ as in Lemma 3.5. The pattern J is exactly the new first-relay pattern for k , and therefore

$$\text{Sur}_{S,Y}(T; r, \Phi) = \text{Sur}_{S,Y}(T_0; r, \Phi) + \mathbb{E}[(\Phi(C_S) - m_k^Y); J]. \quad (70)$$

The centered integral of $\Phi(C_k)$ over $D_Y(k)$ is zero. Splitting $D_Y(k)$ according as k hits T_0 or avoids it gives

$$\kappa = \mathbb{E}[(\Phi(C_k) - m_k^Y); D_Y(k), k \leftrightarrow T_0], \quad (71)$$

$$\text{Sur}_{k,Y}(T_0; r, \Phi) - \kappa = \sum_{x \in T_0} \mathbb{P}(J_x^{\{k\},Y}) (m_k^Y - m_x^Y) \geq 0. \quad (72)$$

The last inequality uses maximality of the rank of k .

Multiply (70) by $\mathbb{P}(E)$. Apply Theorem 3.4 to the first term and Lemma 3.5 to the second. We obtain

$$\mathbb{P}(E) \text{Sur}_{S,Y}(T; r, \Phi) \geq \mathbb{P}(J) (\text{Sur}_{k,Y}(T_0; r, \Phi) - \kappa) \geq 0$$

by (72). In the nondegenerate case $\mathbb{P}(E) > 0$, so the desired surplus is nonnegative. The denominator-free boundary argument and rank-free continuity argument preceding this subsection pass the inequality to arbitrary weights. $\square$

4 The monotone projection

We now turn Theorem 3.7 into a comparison with a fixed relay. Fix $a \in V$. For $K \subseteq V$ with $a \notin K$, let $G \setminus K$ denote the random graph obtained by deleting all pairs with an endpoint in K , and define

$$\phi_a(K) := F(K) - \mathbb{E}F(C_a^{G \setminus K}). \quad (73)$$

Lemma 4.1 (projection facts). *Suppose F is increasing.*

(P1) *The map $K \mapsto \phi_a(K)$ is increasing on $\{K : a \notin K\}$ and has an increasing extension to 2^V .*

(P2) *For every source set S and every event H determined by C_S ,*

$$\mathbb{E}[(F(C_S) - F(C_a)); S \not\leftrightarrow a, H] = \mathbb{E}[\phi_a(C_S); S \not\leftrightarrow a, H]. \quad (74)$$

(P3) *For every vertex x ,*

$$\mathbb{E}[\phi_a(C_x); x \not\leftrightarrow a] = \mathbb{E}F(C_x) - \mathbb{E}F(C_a). \quad (75)$$

Proof. If $K \subseteq L$ and neither contains a , then $F(K) \leq F(L)$, while deleting all pairs meeting L leaves no more open paths from a than deleting all pairs meeting K . Thus the second term in (73) decreases, proving (P1) on the indicated subposet. Give every set containing a the value $\max_{a \notin K} \phi_a(K)$; this is an increasing extension. A final constant shift makes it nonnegative if desired.

On $S \not\leftrightarrow a$, condition on $C_S = K$. Every pair from K to its complement is closed, and the edges wholly outside K retain their independent laws. Consequently the conditional law of C_a is exactly the law of $C_a^{G \setminus K}$. This proves (P2). Apply (P2) with $S = \{x\}$ and H the sure event. On $x \leftrightarrow a$ the two clusters, and hence their F -values, agree. Adding this zero contribution proves (P3). $\square$

The repository already contains the singleton versions:

- `CovTau.monotone_projFun`;
- `CovTau.tower_clusterFun`;
- `CovTau.setIntegral_sub_eq_projFun`.

The proof above shows that the same interface works for the explored cluster of a source set.

5 A fixed minimizer works: Conjecture 4

Theorem 5.1 (fixed-minimizer source-set comparison). *Let $F : 2^V \rightarrow \mathbb{R}$ be increasing, let $A \neq \emptyset$, and choose $a \in A$ such that*

$$\mathbb{E}F(C_a) = \min_{x \in A} \mathbb{E}F(C_x). \quad (76)$$

Then, for every finite source set S ,

$$\mathbb{E}[(F(C_S) - F(C_a)); S \not\leftrightarrow a, S \leftrightarrow A \setminus \{a\}] \geq 0. \quad (77)$$

Consequently

$$\mathbb{E}[F(C_S); S \leftrightarrow A] \geq \mathbb{E}[F(C_a); S \leftrightarrow A]. \quad (78)$$

For $S = \{o\}$, the contribution on $o \leftrightarrow a$ vanishes identically, so (78) holds with equality removed only on the avoided part.

Proof. Put $Y = \{a\}$ and $T = A \setminus \{a\}$. Use the increasing projection ϕ_a from Lemma 4.1. By (75) and (76), every conditional relay mean is nonnegative:

$$\mathbb{E}[\phi_a(C_x) \mid x \not\leftrightarrow a] = \frac{\mathbb{E}F(C_x) - \mathbb{E}F(C_a)}{\mathbb{P}(x \not\leftrightarrow a)} \geq 0. \quad (79)$$

Relays for which the denominator is zero never occur in an a -avoiding first pattern and may be removed. Rank the others by the quantities in (79). Theorem 3.7, followed by Lemma 4.1(P2), gives

$$\begin{aligned} & \mathbb{E}[(F(C_S) - F(C_a)); S \not\leftrightarrow a, S \leftrightarrow T] \\ &= \mathbb{E}[\phi_a(C_S); S \not\leftrightarrow a, S \leftrightarrow T] \\ &\geq \sum_{x \in T} \mathbb{P}(J_x^{S, \{a\}}) \mathbb{E}[\phi_a(C_x) \mid x \not\leftrightarrow a] \geq 0. \end{aligned}$$

This is (77).

On the remaining part of $\{S \leftrightarrow A\}$, the source cluster union meets a . Hence $C_a \subseteq C_S$ and $F(C_a) \leq F(C_S)$. Adding that nonnegative contribution proves (78). $\square$

Corollary 5.2 (Conjecture 4, in a stronger form). *For a singleton source $S = \{o\}$, Theorem 5.1 says that the particular unconditional minimizer a in (76) satisfies*

$$\mathbb{E}[F(C_o); o \leftrightarrow A] \geq \mathbb{E}[F(C_a); o \leftrightarrow A]. \quad (80)$$

Thus Conjecture 4 holds. This also answers the first Kozma–Nitzan Question 7: the minimizer of the post-FKG quantity is a valid relay in the pre-FKG comparison.

6 Conjectures 2, 1, and 3

Corollary 6.1 (strong Conjecture 2). *For all o, b, A ,*

$$\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) \geq \min_{x \in A} \mathbb{P}(x \leftrightarrow b, o \leftrightarrow A). \quad (81)$$

In particular, Conjecture 2 holds.

Proof. Apply Corollary 5.2 to $F(K) = \mathbf{1}\{b \in K\}$. Then $F(C_x) = \mathbf{1}\{x \leftrightarrow b\}$, and (80) is at least as strong as (81). Finally $\mathbb{P}(o \leftrightarrow b) \geq \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A)$. $\square$

Corollary 6.2 (Conjecture 1). *The multiplicative inequality (2) holds for every finite weighted graph.*

Proof. Both $\{o \leftrightarrow A\}$ and $\{x \leftrightarrow b\}$ are increasing events. Harris–FKG and Corollary 6.1 give

$$\begin{aligned} \mathbb{P}(o \leftrightarrow b) &\geq \min_{x \in A} \mathbb{P}(o \leftrightarrow A, x \leftrightarrow b) \\ &\geq \min_{x \in A} \mathbb{P}(o \leftrightarrow A) \mathbb{P}(x \leftrightarrow b) \\ &= \mathbb{P}(o \leftrightarrow A) \min_{x \in A} \mathbb{P}(x \leftrightarrow b). \end{aligned}$$

$\square$

Remark 6.3. Conjecture 1 also follows directly from the repository's unavowed GEN theorem by taking $F(K) = \mathbf{1}\{b \in K\}$ and ordering relays by $\mathbb{P}(x \leftrightarrow b)$. The route above is included to display the implication chain in the Kozma–Nitzan paper.

Corollary 6.4 (Conjecture 3). *Conjecture 3 holds; for $0 < \varepsilon \leq 1$ one may take $\delta = \varepsilon/2$.*

Proof. Under its hypotheses, Corollary 6.2 gives

$$\mathbb{P}(o \leftrightarrow b) > (1 - \delta)^2 = 1 - \varepsilon + \frac{\varepsilon^2}{4} > 1 - \varepsilon.$$

For $\varepsilon > 1$ the assertion is trivial, for example with $\delta = 1/2$. □

Proposition 6.5 (countable-graph closure). *On a countable graph with independent, possibly inhomogeneous bond weights, Conjecture 1 holds with $\inf_{x \in A}$ in place of the minimum when A is infinite. Consequently Conjecture 3 holds there with the same graph-independent choice $\delta = \varepsilon/2$.*

Proof. First let A be finite. Exhaust the countable edge set by finite edge sets, retaining their original weights and declaring every other edge closed. Each connection event between specified vertices increases to the corresponding event in the full graph, because every connecting path is finite. Apply Corollary 6.2 at every finite stage and use continuity from below; the minimum of finitely many convergent sequences converges to the minimum of their limits.

For countably infinite A , exhaust it by nonempty finite sets A_m . Then $\{o \leftrightarrow A_m\}$ increases to $\{o \leftrightarrow A\}$ and $\min_{x \in A_m} \mathbb{P}(x \leftrightarrow b)$ decreases to $\inf_{x \in A} \mathbb{P}(x \leftrightarrow b)$. Passing to the limit proves the first claim. Under the hypotheses of Conjecture 3, continuity from below also provides a finite $A_0 \subseteq A$ with $\mathbb{P}(o \leftrightarrow A_0) > 1 - \delta$. The already proved finite-relay inequality then gives the same estimate as in Corollary 6.4. □

Corollary 6.6 (the main critical-percolation consequence). *For Bernoulli bond percolation on $\mathbb{Z}^d$, $d \geq 2$,*

$$\mathbb{P}_{p_c}(|C(0)| = \infty) = 0.$$

Proof. Proposition 6.5 supplies Conjecture 3 in the uniform form assumed by Theorem 6 of Kozma–Nitzan. That theorem gives the displayed conclusion. □

7 The new edge argument: Conjecture 6

The source-set extension makes the edge-contraction conjecture a short corollary. We isolate the deterministic observation on which it rests.

Lemma 7.1 (an internal edge does not change a source union). *Let $e = \{v, w\}$, let $S = \{v, w\}$, and couple all weightings of e by using the same states for every other pair. Then the following objects do not depend on the state of e :*

- (E1) *the vertex set $C_S = C_v \cup C_w$;*
- (E2) *every event $\{S \leftrightarrow B\}$ and $\{S \nleftrightarrow B\}$;*
- (E3) *on $S \nleftrightarrow a$, the cluster C_a .*

When e is forced open, $C_v = C_w = C_S$.

Proof. Opening an edge with both endpoints in the source set merely joins two components already included in their union; it adds no vertex to that union. This proves (E1), and (E2) follows. On $S \not\leftrightarrow a$, the cluster of a contains neither endpoint of e , so changing e cannot change that cluster. The last assertion is immediate when e is open. $\square$

Theorem 7.2 (fixed-minimizer contraction; stronger than Conjecture 6). *Let $e = \{v, w\}$ and let $a \in A$ minimize $\mathbb{P}_G(x \leftrightarrow b)$ over A . Then, without any endpoint hypotheses,*

$$\mathbb{P}_{G/e}(v \leftrightarrow b, v \leftrightarrow A) \geq \mathbb{P}_{G/e}(a \leftrightarrow b, v \leftrightarrow A), \quad (82)$$

and consequently

$$\mathbb{P}_{G/e}(v \leftrightarrow b) \geq \mathbb{P}_{G/e}(v \leftrightarrow A) \mathbb{P}_{G/e}(a \leftrightarrow b). \quad (83)$$

Proof. Use $F(K) = \mathbf{1}\{b \in K\}$ in Theorem 5.1, in the original weighted graph G , with the source set $S = \{v, w\}$. It gives

$$\mathbb{E}_G[(F(C_S) - F(C_a)); S \not\leftrightarrow a, S \leftrightarrow A \setminus \{a\}] \geq 0. \quad (84)$$

By Lemma 7.1, the random variable and event in (84) are independent of the state of e . We may therefore evaluate them with e forced open. In G/e , the source union C_S is C_v . Moreover, on $v \leftrightarrow a$ the two indicators $F(C_v)$ and $F(C_a)$ agree, while on $v \not\leftrightarrow a$ the event $v \leftrightarrow A$ is the event $v \leftrightarrow A \setminus \{a\}$. It follows that the left side of (84) is exactly

$$\mathbb{P}_{G/e}(v \leftrightarrow b, v \leftrightarrow A) - \mathbb{P}_{G/e}(a \leftrightarrow b, v \leftrightarrow A).$$

This proves (82).

Finally, Harris–FKG in G/e gives

$$\mathbb{P}_{G/e}(a \leftrightarrow b, v \leftrightarrow A) \geq \mathbb{P}_{G/e}(a \leftrightarrow b) \mathbb{P}_{G/e}(v \leftrightarrow A).$$

Together with $\mathbb{P}_{G/e}(v \leftrightarrow b) \geq \mathbb{P}_{G/e}(v \leftrightarrow b, v \leftrightarrow A)$ and (82), this proves (83). $\square$

Corollary 7.3 (Conjecture 6). *Conjecture 6 holds. Its two assumptions concerning v and w are superfluous.*

8 The numbered questions: common coefficients and candidate relays

We now dispose of Questions 5 and 7–9. The coefficient conclusion below is stronger than the one printed in the paper.

Theorem 8.1 (common coefficients; Question 5). *For fixed G, o, A there are coefficients $c_x \geq 0$, $x \in A$, with $\sum_{x \in A} c_x = 1$, such that simultaneously for every $b \in V$,*

$$\sum_{x \in A} c_x \mathbb{P}(o \leftrightarrow A, x \leftrightarrow b) \leq \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) \leq \mathbb{P}(o \leftrightarrow b). \quad (85)$$

In particular, Question 5 has an affirmative answer.

Proof. Let $E = \{o \leftrightarrow A\}$ and regard the following as vectors indexed by $b \in V$:

$$r_b = \mathbb{P}(E, o \leftrightarrow b), \quad q_b^x = \mathbb{P}(E, x \leftrightarrow b) \quad (x \in A).$$

For every vector $\lambda = (\lambda_b)_{b \in V} \geq 0$, the functional

$$F_\lambda(K) = \sum_{b \in K} \lambda_b$$

is nonnegative and increasing. Conjecture 4, already proved in Corollary 5.2, gives

$$\lambda \cdot r = \mathbb{E}[F_\lambda(C_o); E] \geq \min_{x \in A} \mathbb{E}[F_\lambda(C_x); E] = \min_{x \in A} \lambda \cdot q^x. \quad (86)$$

Put $K = \text{conv}\{q^x : x \in A\} \subseteq \mathbb{R}^V$ and $D = r - \mathbb{R}_{\geq 0}^V$. Suppose $K \cap D = \emptyset$. Strict separation of the compact convex set K from the closed convex set D then gives a nonzero vector λ such that

$$\inf_{z \in K} \lambda \cdot z > \sup_{y \in D} \lambda \cdot y.$$

Since D is unbounded in every negative coordinate direction, finiteness of the supremum forces $\lambda \geq 0$, and that supremum is $\lambda \cdot r$. The infimum over K is $\min_{x \in A} \lambda \cdot q^x$, contradicting (86). Hence $K \cap D \neq \emptyset$. Write a point in the intersection as $\sum_{x \in A} c_x q^x = r - u$, where $c_x \geq 0$, $\sum_x c_x = 1$, and $u \geq 0$. Coordinatewise this is the first inequality in (85); the second is event inclusion. The single vector (c_x) was chosen using all coordinates at once and is therefore independent of b . $\square$

Corollary 8.2 (Questions 7 and 8). *Questions 7 and 8 both have affirmative answers, including every choice among tied minimizers.*

Proof. For Question 7, take $F(K) = \mathbf{1}\{b \in K\}$ in Corollary 5.2: the specified minimizer of $\mathbb{P}(x \leftrightarrow b)$ satisfies (7). For Question 8, the specified a realizes $\min_{x \in A} \mathbb{P}(o \leftrightarrow A, x \leftrightarrow b)$, so the strong form (81) of Conjecture 2 is exactly (7). $\square$

Theorem 8.3 (deleted-star minimizer; Question 9). *Let H be obtained from G by deleting every pair incident to o , and let $a \in A$ minimize $\mathbb{P}_H(x \leftrightarrow b)$ over $x \in A$. Then (7) holds in G . Thus Question 9 has an affirmative answer, again for every tied minimizer.*

Proof. Expose all pairs incident to o . For a fixed star configuration η , let

$$N_\eta = \{u \neq o : \{o, u\} \text{ is open in } \eta\}, \quad S_\eta = \{o\} \cup N_\eta.$$

The unexposed pairs have exactly the percolation law of H , and, pointwise in their configuration,

$$C_o^G = C_{S_\eta}^H, \quad \{o \leftrightarrow_G A\} = \{S_\eta \leftrightarrow_H A\}. \quad (87)$$

The inclusion of the isolated vertex o in S_η makes these formulas valid without exceptions when $o \in A$ or $b = o$.

Put $F_b(K) = \mathbf{1}\{b \in K\}$. On $S_\eta \leftrightarrow_H a$, the full G -clusters of o and a coincide, so the two integrands in (7) agree. On $S_\eta \not\leftrightarrow_H a$, no open star pair can change C_a , whence $C_a^G = C_a^H$; moreover, on this event a hit of A is a hit of $A \setminus \{a\}$. Therefore the difference between the two sides of (7) has the exact decomposition

$$\begin{aligned} & \mathbb{P}_G(o \leftrightarrow b, o \leftrightarrow A) - \mathbb{P}_G(a \leftrightarrow b, o \leftrightarrow A) \\ &= \sum_{\eta} \mathbb{P}(\eta) \mathbb{E}_H[(F_b(C_{S_\eta}) - F_b(C_a)); S_\eta \not\leftrightarrow a, S_\eta \leftrightarrow A \setminus \{a\}]. \end{aligned} \quad (88)$$

The minimizer assumption is $\mathbb{E}_H F_b(C_a) \leq \mathbb{E}_H F_b(C_x)$ for every $x \in A$. Apply Theorem 5.1 in the graph H to each deterministic source set S_η . Every expectation on the right of (88) is nonnegative. Averaging proves the theorem. This also handles source–relay overlaps and star configurations of zero probability; no division is used. $\square$

9 Dependency graph and audit checklist

The logical chain is

$$\boxed{\text{CSH}_{\text{set source}}} \implies \boxed{\text{SAG}} \implies \boxed{\text{fixed-minimizer C4}} \implies \boxed{\text{C4} \Rightarrow \text{C2} \Rightarrow \text{C1} \Rightarrow \text{C3}},$$

and independently

$$\boxed{\text{fixed-minimizer C4 for } S = \{v, w\}} \implies \boxed{\text{fixed-minimizer pre-FKG in } G/e} \implies \boxed{\text{C6}}.$$

The remaining numbered questions branch off as follows:

$$\boxed{\text{C4}} \implies \boxed{\text{Q5}}, \quad \boxed{\text{fixed-minimizer C4}} \implies \boxed{\text{Q7}}, \quad \boxed{\text{strong C2}} \implies \boxed{\text{Q8}},$$

and

$$\boxed{\text{fixed-minimizer C4 for arbitrary } S} + \boxed{\text{deleted-star conditioning}} \implies \boxed{\text{Q9}}.$$

For auditing, the sign-sensitive steps are only these:

1. The set-source Markov exploration gives $-\widehat{\Phi}(C_R)$, not $-\widehat{\Phi}(C_d)$. Their difference is the non-negative error (38); because the level split subtracts the centred bracket, that error enters (50) with a plus sign.
2. Every evaluation made by a tail level form or the final observer is a singleton. Hence (40), rather than a positivity-preservation assertion for a signed level operator, is what preserves the preceding inequality.
3. In the nonempty-intersection proof of (27), the cross-deleted sets P, P' are essential: they make the arguments exactly $s \cup t$ and $s \cap t$ before applying four-functions.
4. In (72), the peeled relay is rank *maximal*, so $m_k^Y - m_x^Y \geq 0$.
5. In Lemma 3.5, $C_k \subseteq C_S$ and Φ is increasing, so replacing $\Phi(C_S)$ by $\Phi(C_k)$ is a lower bound.
6. The residual probability $q(K)$ in that lemma is increasing: a larger explored cluster deletes more possible paths to the forbidden set.
7. In the projection, $K \subseteq L$ makes the first term increase and the conditional a -cluster term decrease.
8. The disjoint transfer theorem is used only when $S \cap (Y \cup T) = \emptyset$. If the source already contains a relay, Lemma 3.6 supplies the missing branch.
9. In the edge step one must restrict first to $S \not\leftrightarrow a$. Without that restriction, C_a can change when e is opened.
10. In Question 5, the separating vector is forced into the positive orthant because $r - \mathbb{R}_{\geq 0}^V$ is unbounded in every negative coordinate direction. This is what permits the use of an increasing linear cluster functional.

11. In Question 9, the random source is $S_\eta = \{o\} \cup N_\eta$, not merely N_η . Retaining o covers $o \in A$, $a = o$, and $b = o$ without exceptional cases.

No limiting argument chooses a minimizer. This is important at weights 0 or 1 and in the presence of ties: the proof fixes the specified minimizer a throughout.

10 Lean formalization plan

This section gives declarations at a granularity suitable for parallel formalization. Names are proposals; the mathematical interfaces are the important part.

Block A: source clusters and event identities

Reuse `CovTau.sC` from `Percolation/Continuity/CovTau/A2Defs.lean`, or add a measure-level wrapper `openClusterSet` with mathematical specification

$$\text{openClusterSet}(\omega, S) := \{x : \exists s \in S, \text{Reachable}_\omega(s, x)\}.$$

Prove:

- singleton and union formulas;
- monotonicity in the configuration and in S ;
- the pattern partition (11);
- `clusterSet_pair_independent_internalEdge`, which is Lemma 7.1 in pointwise form.

Block B: avoided means and source surplus

Define denominator-free numerators first:

$$M_x^Y = \mathbb{E}[\Phi(C_x); x \not\rightarrow Y], \quad d_x^Y = \mathbb{P}(x \not\rightarrow Y).$$

Define `sourceFirstPattern` and `sourceAvoidSurplus`. Keep the rank compatibility assumption in cross-multiplied form

$$M_x^Y d_z^Y \leq M_z^Y d_x^Y$$

to avoid division at the boundary of the weight cube.

Block C: guarded CSH and corrected unfolding

Formalize (16)–(19) first in denominator-free form. Make the whole-source exploration identity (37), the correction (38), and the two facts (39)–(40) separate lemmas. Prove the cross-multiplied one-decoy identity (42) before the recursive unfolding (50). This order prevents accidental reuse of the false identity with C_d in place of C_R .

Package the nonempty-intersection proof of (27) as a strong induction on the residual vertex set, exposing the cross-deleted sets P, P' in its four-functions input. The two-half Gibbs step should call the existing antitonicity and conditional-association declarations before closing the regeneration contraction. The output is Theorem 3.1, first at nondegenerate weights and then in a cross-multiplied boundary form.

Block D: set-source surplus transfer

Generalize the outer observer in `CSH.surplusMargin_nonneg_of_csh`. The required repository pieces already accept sets:

role	existing declaration/file
set cluster	<code>CovTau.sC</code> , <code>A2Defs.lean</code>
set marker covariance	<code>CovTau.BfS</code> , <code>StarBridgeS.lean</code>
two-source horizontal inequality	<code>CovTau.a2H</code> , <code>CovTau.p1H_univ</code>
two-set conditional association	<code>BHK2006_twoSetConditionalAssociation</code>
four-product exchange	<code>setTwoClusterExchange</code>
ordinary CSH peeling	<code>CSH/Peel.lean</code>

The target declaration is the denominator-free form of (57). The proof follows the seven labelled stages in `CSH/Peel.lean`; insert Y into every avoided set and use $C_z \subseteq C_S$ at the source evaluation.

Block E: top-relay estimate and SAG

Formalize Lemma 3.5 separately. The Markov/tower step can reuse the set-cluster exploration API; the monotonicity of $q(K)$ should be its own lemma. Then formalize the four displayed equations (70)–(72) and close the induction with Block C. Export both the ratio form (69) (under positive denominators) and a cross-multiplied boundary form.

Block F: set-source projection

Generalize `CovTau.monotone_projFun` and `CovTau.setIntegral_sub_eq_projFun` from a singleton explored cluster to `openClusterSet`. Prove precisely (74) and (75).

Block G: common coefficients

Formalize Theorem 8.1 as a finite-dimensional alternative: either a point of $\text{conv}\{q^x : x \in A\}$ lies below r coordinatewise, or a nonnegative separating functional contradicts Conjecture 4 applied to F_λ . A Mathlib theorem of alternatives may replace strict separation, but the output must be one simplex vector shared by every target b .

Block H: deleted-star conditioning

Represent the incident-pair coordinates as a finite product and prove (87) pointwise for each star state η . Split on $S_\eta \leftrightarrow_H a$, prove the exact finite-sum identity (88), and apply the arbitrary-source version of Theorem 5.1. State the source as $S_\eta = \{o\} \cup N_\eta$ to cover all overlap and endpoint cases.

Block I: final theorem statements

Add named propositions for:

- `kozmaNitzan_conjecture4_fixedMin`;
- `kozmaNitzan_conjecture2_strong`;
- `kozmaNitzan_conjecture1`;

- the existing Conjecture 3 theorem as a corollary;
- `kozmaNitzan_conjecture6_strong`, whose statement omits the endpoint assumptions.
- `kozmaNitzan_question5_commonCoefficients`;
- `kozmaNitzan_question8`;
- `kozmaNitzan_question9`.

The recommended kernel audit is

```
#print axioms kozmaNitzan_conjecture6_strong
```

followed by the corresponding commands for Conjectures 4, 2, and 1. Question 7 is already represented by `KNA11.question7_holds`; Question 8 should be exported as a named one-line corollary of the strong Conjecture 2 declaration.

11 Conclusion

The missing idea is not another estimate for a single observer. It is to retain the entire explored union C_S when several possible observer components are present. The CSH proof was already built from set-cluster exploration and two-set conditional association, so the source-set lift preserves its induction. With $S = \{o\}$ it selects the unconditional minimizer in the pre-FKG comparison. With $S = \{v, w\}$ it makes the state of the edge $\{v, w\}$ internal to the source union, which is exactly the invariance needed to force that edge open. This proves all five numbered Kozma–Nitzan conjectures. Convex separation, the fixed-minimizer specialization, and deleted-star conditioning then answer Questions 5 and 7–9. Hence every numbered conjecture and question in the supplied Kozma–Nitzan paper is resolved within the finite weighted-graph model. The countable-graph closure then feeds Conjecture 3 into their reduction and yields the paper’s stated critical-percolation consequence.

References

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